


# Türkiye dossier summary

*Irrigation water, climate, fertilisers and crops. The essentials of the 25-page dossier.*

## Key figures

| Figure | What it means |
|---|---|
| **8.63 bn m³** | Water saved per year in the central scenario, or 15.4% of agricultural withdrawals. Range of 4.3 to 17.3 billion m³ a year. |
| **756 000 ha** | Extra irrigable area with the same water, at 11,404 m³ per irrigated hectare. |
| **6.82 Mt** | Production gain in the central scenario, for the crops where a trial is cited. |
| **4.91 M ha** | Area actually irrigated in 2024 (FAOSTAT); DSİ puts the irrigable area at 8.5 million hectares. |
| **56.0 km³** | Water withdrawn by agriculture (2022), or 87% of total withdrawals. |
| **30.7 bn$** | Agricultural exports in 2024, including $1,503 million of shelled hazelnuts; imports: $23.1 bn. |
| **1.92 Mt** | Mineral nitrogen used in agriculture (FAOSTAT, latest year). |

*Bn = billion · M = million · Mm³ = million cubic metres · t = tonnes. The water savings are scenarios built from the trials cited.*

## Our products

Green Solutions develops products for water, fertilisation and crop protection. Four families are of interest to this dossier.

| Product | Claimed use | Published evidence (level) |
|---|---|---|
| **EVERGREEN® (water-retention polymer)** | Holds water as a gel, near the roots. | Trials on maize (Türkiye), potato, watermelon, date palm and soybean. Level B to C. |
| **ECOSORB® (water-retention polymer based on corn starch)** | Same use; lasting 2 to 3 years according to the website. | Soybean trial (China); laboratory acrylamide analysis. Level B to C. |
| **ECOFERT® (fertiliser)** | Fertiliser certified by SOHISCERT and EU Bio. | Published certificate; potato trial with EVERGREEN®. Levels A and C. |
| **Paulownia ALTIFOLIA®** | Fast-growing tree for agroforestry. | Website figures; no trial in Türkiye. Level C. |

## Key points

- **Wheat dominates the land.** 6.93 million hectares harvested in 2024, almost always without irrigation.
- **Maize has its trial in Türkiye itself.** At the TIGEM state farm in Konya (2025): Irrigation −38.9%, yield +17.1%. Maize accounts for 20% of the central saving.
- **Vegetables give the largest saving.** 2,524 Mm³ a year in the central scenario, on 732 000 ha.
- **A very contrasted climate.** Konya, on the central plateau, gets three times less rain than Bursa or Samsun; the Southeast has the largest water deficit.

- **Importing from Switzerland.** 0% customs duty for goods originating in EFTA (6.5% under the general tariff); 20% VAT; registration with the Ministry of Agriculture (BÜGEM).

## The regions at a glance

| Region (city) | Rainfall (mm/yr) | Deficit (mm/yr) | Temperature (°C) |
|---|---|---|---|
| Thrace (Edirne) | 540 | 606 | 14.1 |
| Marmara (Bursa) | 861 | 177 | 15.0 |
| Aegean (Manisa) | 698 | 609 | 16.9 |
| Mediterranean (Adana) | 718 | 615 | 18.4 |
| Central Anatolia (Konya) | 299 | 1,029 | 12.5 |
| Black Sea (Samsun) | 856 | 68 | 13.8 |
| Eastern Anatolia (Erzurum) | 624 | 369 | 5.8 |
| Southeastern Anatolia (Şanlıurfa) | 491 | 1,181 | 18.2 |

## How to read the figures

Each figure refers to a source with a confidence level. A: Official statistics or international database (World Bank, FAO, Turkish administration). B: Research institute, laboratory or public body outside statistics. C: Manufacturer statement, website, press or estimate. The sources are listed in Annex A of the dossier.

## Water saved by crop group (central scenario)

| Crop group | Irrigated area (ha) | Water saved ($Mm^3$/yr) | Trial or assumption |
|---|---|---|---|
| Vegetables and melons | 731,538 | 2,524 | Watermelon, Mexico (2023) |
| Grain maize | 789,379 | 1,755 | Maize, Türkiye (TIGEM, 2025) |
| Orchards and vineyards (apple, apricot, citrus, cherry, grape…) | 1,140,917 | 1,301 | Generic assumption |
| Other irrigated crops (wheat, sunflower, fodder, olives…) | 1,026,900 | 1,171 | Generic assumption |
| Potato and onion | 261,437 | 775 | Potato, South Africa (2025) |
| Cotton | 466,927 | 533 | Generic assumption |
| Sugar beet | 324,286 | 370 | Generic assumption |
| Rice (paddy) | 128,904 | 147 | Generic assumption |
| Soybean | 43,712 | 50 | Yield: Soybean, China |

## Trade

- **First products exported.** Shelled hazelnuts ($1,503 million), crude sunflower oil and wheat flour, out of $30.7 bn of agricultural exports.
- **Main imports.** Soybeans ($1,766 million), cotton lint, wheat and crude sunflower oil, for the processing industry.

*Full dossier: 25 pages, with the sources (Annex A) and the calculation method (Annex B). Green Solutions · contact@green-solutions-group.com · green-solutions-group.com*